
```
clear
clc
close all
```

Problem 4

```
%{
Discrete Steps:
U: 1/2
I: 1/4, 1/2, 1/4 --> 1/4, 3/4
M: 1/4, 3/4 --> 1/4
N: 3/4, 1/4 --> 1/4
N': 1/2
A: 1
%}
n=10e4;
y_u=0;
t_u=zeros(n,1);
t_u_A=[];
t_u_F=[];
for i=1:n

    U=rand(1);
    [x,t_u(i)]=discrete_U(U,t_u(i));
    y_u=y_u+x; %counts number of folded

    if x==1
        t_u_F(end+1,1)=t_u(i); %takes values for x=1, folded
    else
        t_u_A(end+1,1)=t_u(i); %takes values for x=0, aggregate
    end
end
p_u=y_u./n
tau_U_F=mean(t_u_F);
tau_U_A=mean(t_u_A);
tau_U=mean(t_u);

t_i_F=[];
t_i_A=[];
y_i=0;
t_i=zeros(n,1);
for i=1:n

    U=rand(1);

    [x,t_i(i)]=discrete_I(U,t_i(i));
    y_i=y_i+x;
    if x==1
        t_i_F(end+1,1)=t_i(i);
    else
        t_i_A(end+1,1)=t_i(i);
    end
end
%}
```

```

        end

    end

    p_i=y_i./n
    tau_i_F=mean(t_i_F);
    tau_i_A=mean(t_i_A);
    tau_i=mean(t_i);
    figure
    hold on
    histogram(t_i_F, 'Normalization', 'pdf')
    xline(tau_i_F)
    xlabel('Number of steps folded')
    ylabel('Probability')
    legend('Histogram','Tau_x')
    title('Distribution of Steps Starting at I Folded')
    hold off
    figure
    histogram(t_i_A, 'Normalization', 'pdf')
    xline(tau_i_A)
    xlabel('Number of steps aggregate')
    ylabel('Probability')
    legend('Histogram','Tau_x')
    title('Distribution of Steps Starting at I Aggregate')
    y_m=0;
    t_m=zeros(n,1);
    t_m_F=[];
    t_m_A=[];
    for i=1:n

        U=rand(1);
        [x,t_m(i)]=discrete_M(U,t_m(i));
        y_m=y_m+x;

        if x==1
            t_m_F(end+1,1)=t_m(i);
        else
            t_m_A(end+1,1)=t_m(i);
        end
    end

    p_m=y_m./n
    tau_M_F=mean(t_m_F);
    tau_M_A=mean(t_m_A);
    tau_m=mean(t_m);
    %Functions to retrieve aggregate or folded value using discrete inverse
    %transform
    function [y,t]=discrete_U(U,t)
    V=rand(1);
    t=t+1;
    if U<=.5
        [y,t]=discrete_I(V,t);
    else
        [y,t]=discrete_M(V,t);
    end
end
end

```

```

function [y,t]=discrete_M(U,t)
V=rand(1);
t=t+1;
if U<=.25
    y=0;
else
    [y,t]=discrete_U(V,t);
end
end
function [y,t]=discrete_I(U,t)
V=rand(1);
t=t+1;
if U<=.5
    y=1;
elseif U<=.75
    y=0;
else
    [y,t]=discrete_U(V,t);
end
end

%Creating table
tau_empirical=[tau_U; tau_i; tau_m];
tau_empirical_A=[tau_U_A; tau_i_A; tau_M_A];
tau_empirical_F=[tau_U_F; tau_i_F; tau_M_F];
tau_analytical=[4; 2; 4];

h_empirical=[p_u; p_i; p_m];
h_analytical=[0.5; 0.625; 0.375];
Starting_State=['U'; 'I'; 'M'];
var_name=['State', 'Tau_x', 'Tau_x^F', 'Tau_x^A', 'g_x', 'h_x empirical',
'h_x analytical'];

table(Starting_State, tau_empirical,tau_empirical_F,tau_empirical_A ,
tau_analytical, h_empirical, h_analytical, 'VariableNames', {'State',
'Tau_x', 'Tau_x^F', 'Tau_x^A', 'g_x', 'h_x empirical', 'h_x analytical'})

p_u =

    0.5002

p_i =

    0.6245

p_m =

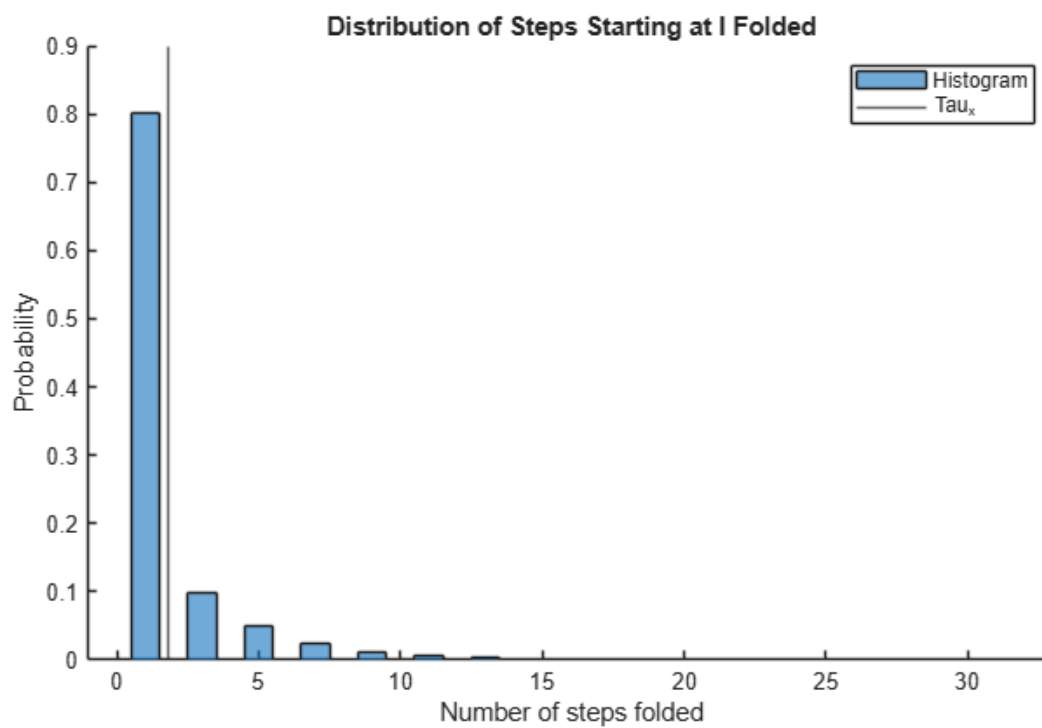
    0.3756

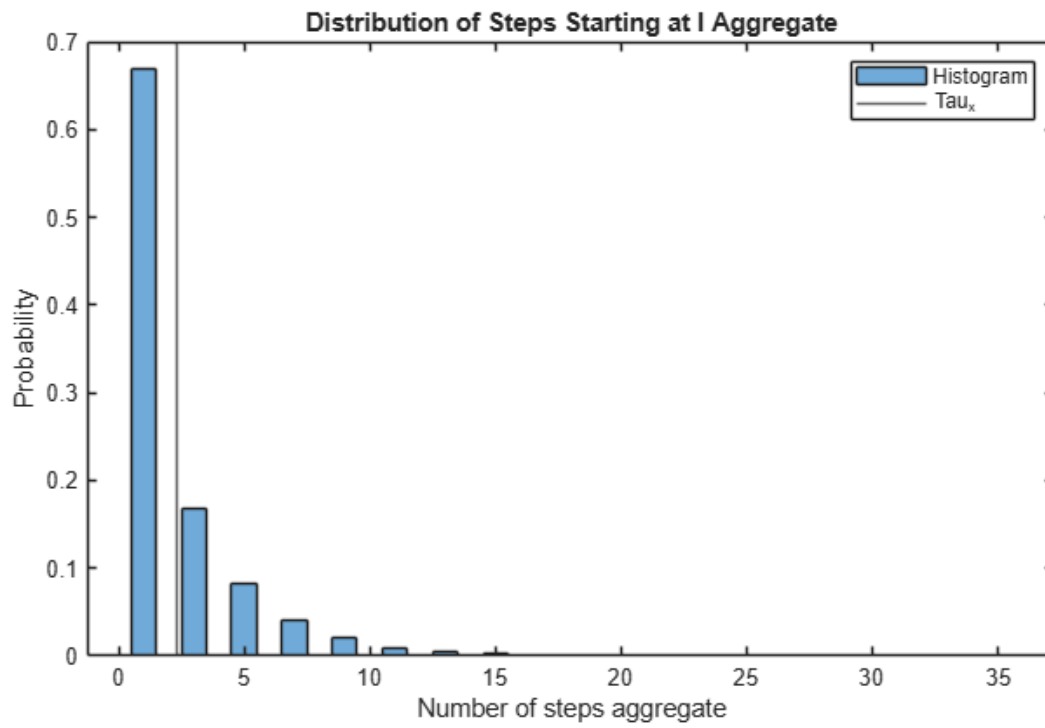
ans =

```

3x7 table

<i>State</i> <i>analytical</i>	<i>Tau_x</i>	<i>Tau_x^F</i>	<i>Tau_x^A</i>	<i>g_x</i>	<i>h_x empirical</i>	<i>h_x</i>
_____	_____	_____	_____	_____	_____	
<i>U</i>	3.9868	3.9954	3.9782	4	0.50017	
0.5 <i>I</i>	1.995	1.8024	2.3154	2	0.62452	
0.625 <i>M</i>	4.0021	5.0013	3.4012	4	0.37556	
0.375						





Published with MATLAB® R2026a